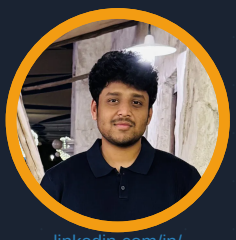

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HANDS-ON AWS PROJECT

Establishing VPC Peering Between Two Public VPCs on AWS

Creating two isolated VPCs with public subnets, establishing a VPC Peering Connection, configuring route tables on both sides, setting up Security Groups, attaching an Elastic IP to Server 1, and verifying cross-VPC connectivity with a live ping test.

2 VPCs	**VPC Peering**	**2 Route Tables**	**Internet Gateway**
2 Security Groups	**2 EC2 Instances**	**Verified via Ping**	

Project Overview

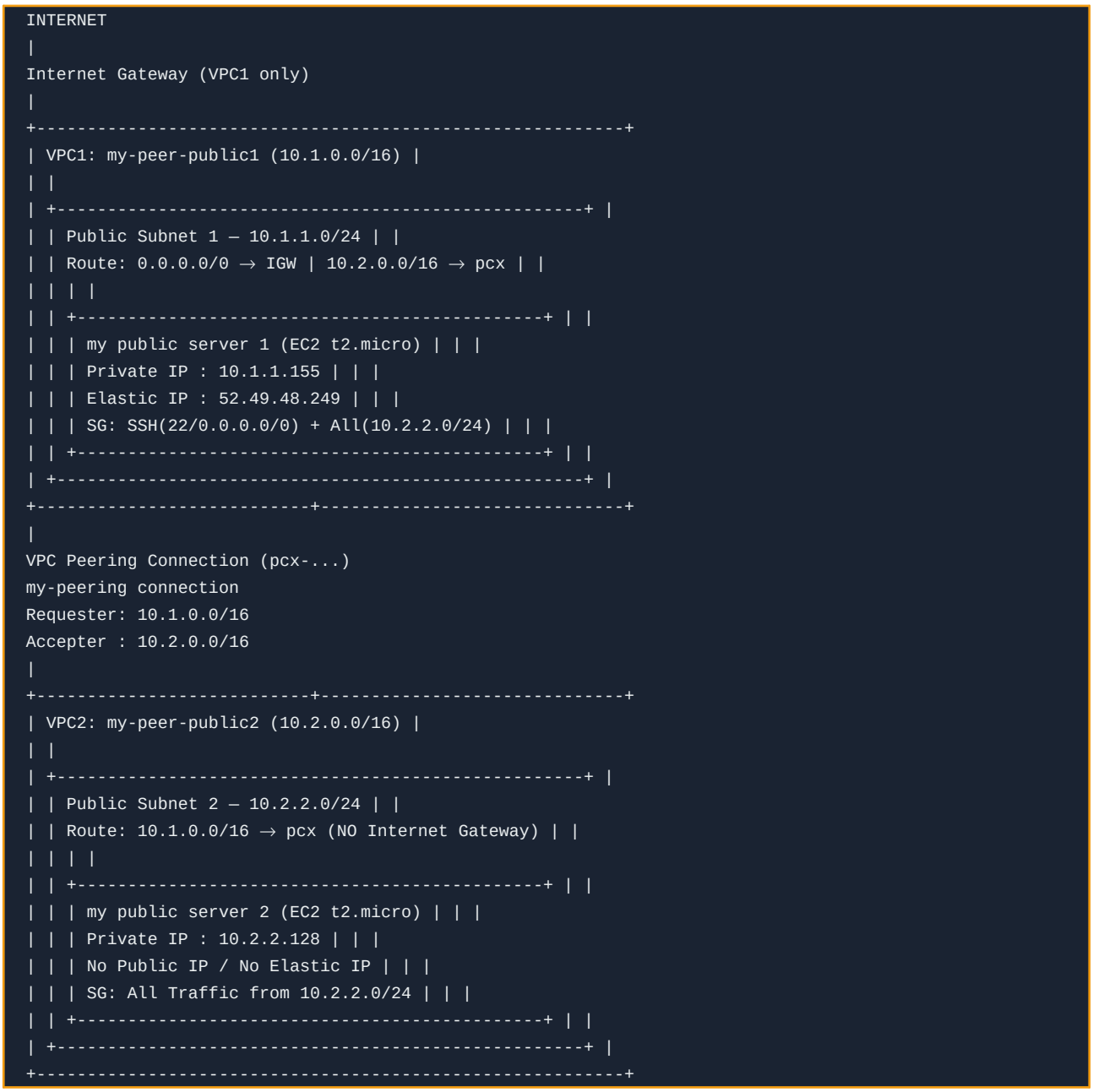
The goal of this project was to connect two isolated AWS VPCs using VPC Peering — simulating a real-world scenario where two separate network environments need to communicate securely. **my-peer-public1** (CIDR **10.1.0.0/16**) acts as the requester VPC with an Internet Gateway and an Elastic IP on its EC2 instance. **my-peer-public2** (CIDR **10.2.0.0/16**) acts as the accepter VPC — it has no Internet Gateway and its EC2 instance has no public IP. Route tables on both VPCs route the peer CIDR through the peering connection. Security Groups enforce access control at the instance level. The design is validated with a live **ping test**.

Component	Name	Configuration
VPC 1	my-peer-public1	10.1.0.0/16
VPC 2	my-peer-public2	10.2.0.0/16
Public Subnet 1	my-peer-publicsubnet1	10.1.1.0/24 (in VPC1)
Public Subnet 2	my-peer-publicsubnet2	10.2.2.0/24 (in VPC2)
Peering Connection	my-peering connection	Active — VPC1 ↔ VPC2
Internet Gateway	Attached to VPC1 only	0.0.0.0/0 → IGW (VPC1 route table)
Route Table 1	my-peer-routes-public1	0.0.0.0/0→IGW | 10.2.0.0/16→pcx
Route Table 2	my-peer-routes-public2	10.1.0.0/16→pcx (no IGW)
Security Group 1	my-peer-public1-sg	SSH 22 (0.0.0.0/0) + All Traffic (10.2.2.0/24)
Security Group 2	my-peer-public2-sg	All traffic from VPC1 subnet
EC2 – Server 1	my public server 1	Private: 10.1.1.155 | Elastic IP: 52.49.48.249
EC2 – Server 2	my public server 2	Private: 10.2.2.128 | No Public IP

1 Architecture Overview

VPC Peering Topology

Two isolated VPCs are connected via a **VPC Peering Connection**. **VPC1** has an Internet Gateway — its EC2 instance has an Elastic IP for external SSH access. **VPC2** has no Internet Gateway — its EC2 instance is reachable only through the peering link from VPC1. Both route tables are updated to route each other's CIDR through the peering connection.

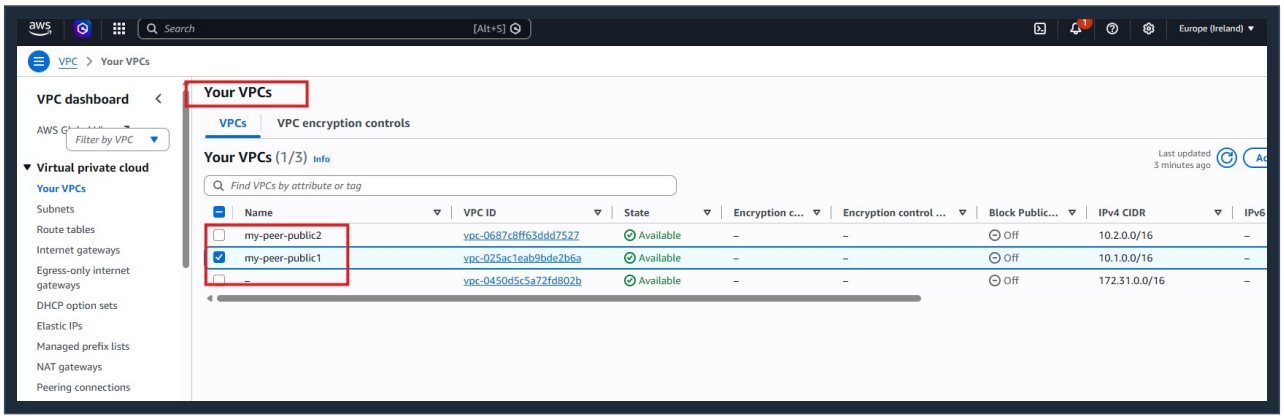


2 Network Foundation

VPCs · Subnets

STEP 01 Create Two VPCs

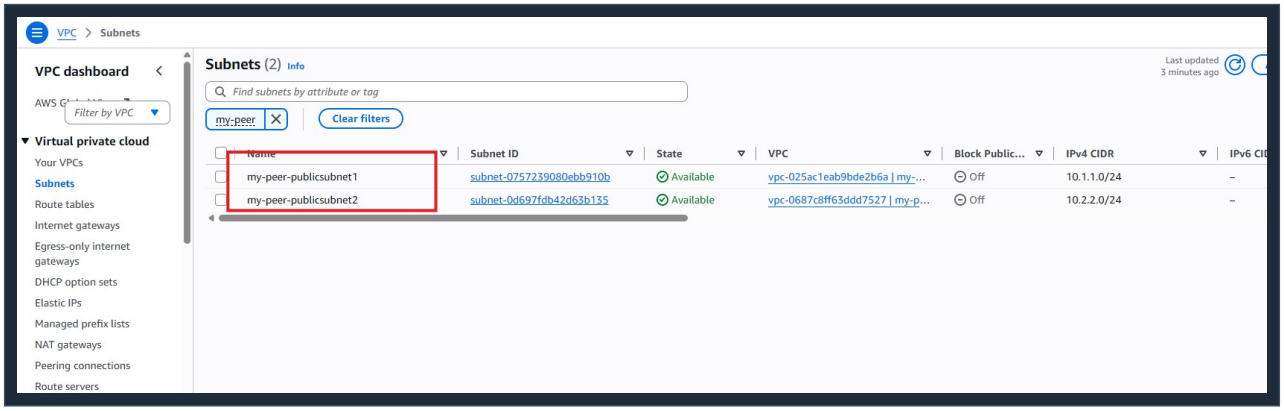
Two custom VPCs were created in the **eu-west-1 (Ireland)** region. **my-peer-public1** (CIDR **10.1.0.0/16**) is the requester VPC. **my-peer-public2** (CIDR **10.2.0.0/16**) is the accepter VPC. Non-overlapping CIDRs are mandatory for VPC Peering.



my-peer-public1 | 10.1.0.0/16 | my-peer-public2 | 10.2.0.0/16 | Available

STEP 02 Create Public Subnets

my-peer-publicsubnet1 (10.1.1.0/24) belongs to VPC1. **my-peer-publicsubnet2** (10.2.2.0/24) belongs to VPC2. Both are public subnets — the difference in internet reachability comes from the route tables, not the subnet type.



my-peer-publicsubnet1 | 10.1.1.0/24 | my-peer-publicsubnet2 | 10.2.2.0/24

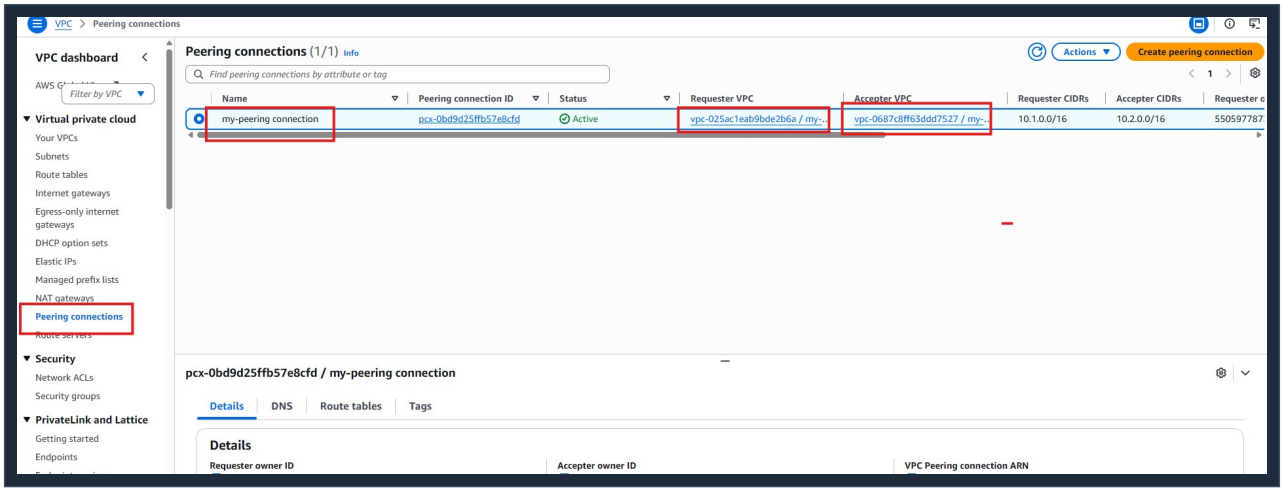
3 VPC Peering Connection

Requester → Accepter → Active

STEP 03 Create & Accept Peering Connection

A VPC Peering Connection named **my-peering connection** was initiated from **my-peer-public1** to **my-peer-public2**. Since both VPCs belong to the same AWS account and region, the request was accepted immediately. Status: **Active**. Requester CIDR: 10.1.0.0/16, Accepter CIDR: 10.2.0.0/16.

■■ **Note:** Owner IDs visible in the AWS console have been omitted from this report for security purposes.



my-peering connection | Active | 10.1.0.0/16 ↔ 10.2.0.0/16

4 Routing Configuration

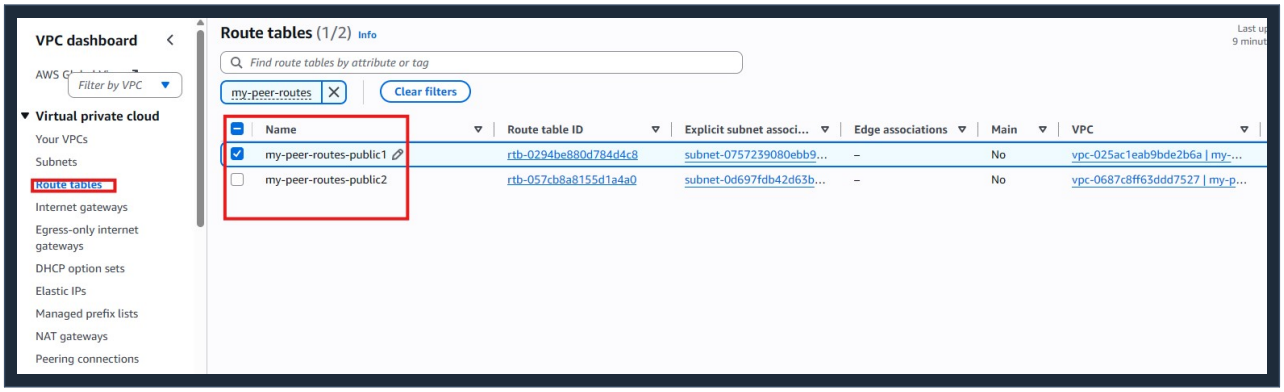
Route Tables · Internet Gateway

STEP 04 **Update Route Tables for Peering + IGW**

Both route tables were updated to enable cross-VPC communication:

my-peer-routes-public1 (VPC1):
• **0.0.0.0/0** → **Internet Gateway** — gives Server 1 internet access
• **10.2.0.0/16** → **pcx (peering connection)** — routes VPC2 traffic through peering

my-peer-routes-public2 (VPC2):
• **10.1.0.0/16** → **pcx (peering connection)** — routes VPC1 traffic back through peering
• **No Internet Gateway route** — VPC2 remains isolated from the internet



Route Table	Destination	Target	Notes
my-peer-routes-public1	0.0.0.0/0	Internet Gateway	VPC1 internet access
my-peer-routes-public1	10.2.0.0/16	pcx (peering)	VPC1 → VPC2
my-peer-routes-public2	10.1.0.0/16	pcx (peering)	VPC2 → VPC1
my-peer-routes-public2	—	No IGW	VPC2 internet isolated

0.0.0.0/0 → IGW | 10.2.0.0/16 → pcx | 10.1.0.0/16 → pcx | No IGW for VPC2

5 Security Groups

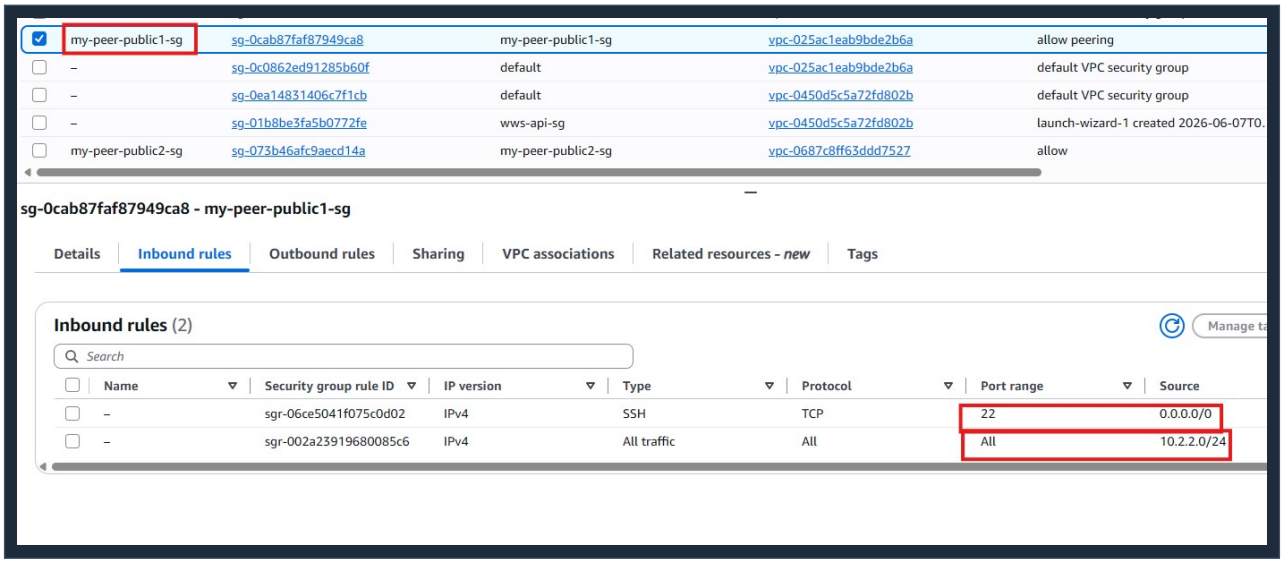
Instance-level stateful firewall rules

STEP 05 Security Group — my-peer-public1-sg (VPC1)

my-peer-public1-sg is attached to the EC2 instance in VPC1. Inbound rules:

• **SSH (Port 22)** — Source: **0.0.0.0/0** *(■■ Open to all — for testing only, not production-safe)*

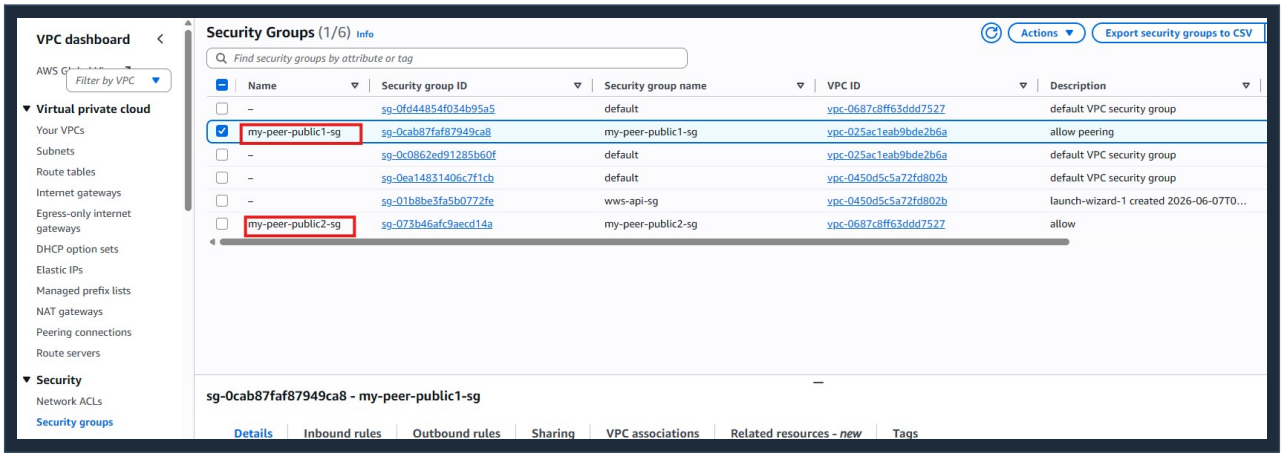
• **All Traffic** — Source: **10.2.2.0/24** (VPC2's subnet CIDR) — allows all inbound traffic arriving via the peering connection from Server 2's subnet



SSH Port 22 | 0.0.0.0/0 | All Traffic | 10.2.2.0/24

STEP 06 Security Groups List — Both VPCs

Two custom security groups are visible: **my-peer-public1-sg** (VPC1, description: 'allow peering') and **my-peer-public2-sg** (VPC2, description: 'allow'). Each is scoped to its own VPC so traffic is controlled independently.



6 Compute & Connectivity

EC2 Instances · Elastic IP

STEP 07 Launch Two EC2 Instances

Two **t2.micro Amazon Linux 2023** instances were launched — one in each VPC's subnet. **my public server 1** is in eu-west-1a (VPC1). **my public server 2** is in eu-west-1b (VPC2) with no public IP. Both show 2/2 status checks passed.

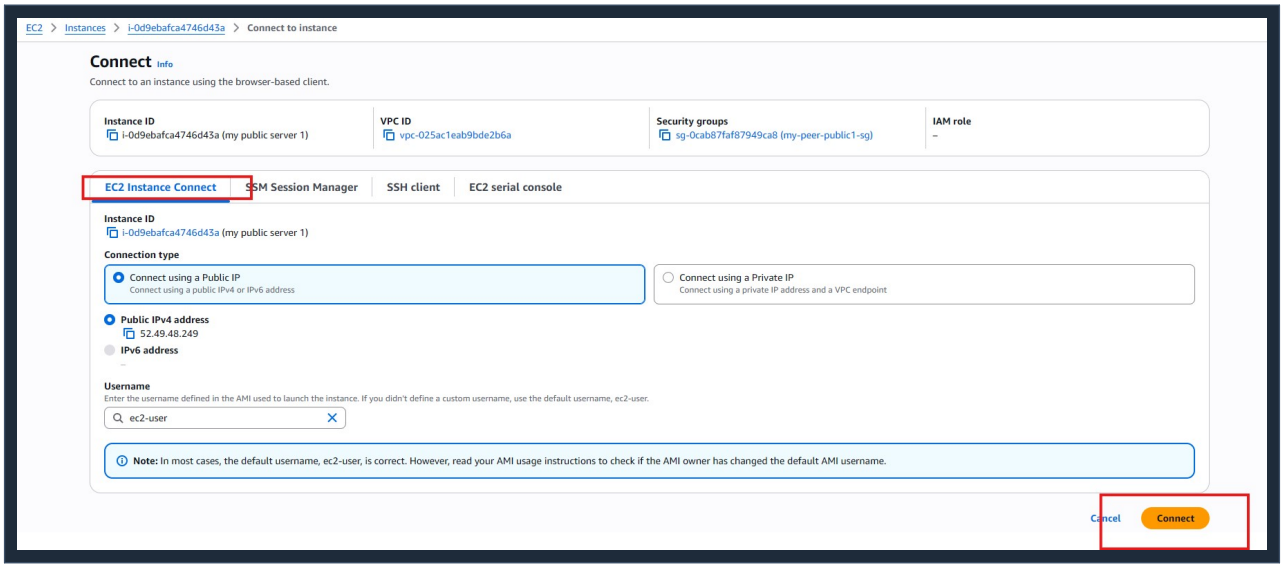
STEP 08 Allocate & Attach Elastic IP to Server 1

Elastic IP **52.49.48.249** was allocated and associated with **my public server 1** (i-0d9ebafca4746d43a). This ensures a persistent public IP regardless of instance restarts. Server 1's private IP remains **10.1.1.155**.

Elastic IP: 52.49.48.249 | Private: 10.1.1.155 | Server 2: 10.2.2.128 | No Public IP – Server 2

STEP 09 Connect to Server 1 via EC2 Instance Connect

Server 1 was accessed using **EC2 Instance Connect** (browser-based SSH). Connection type: Public IP (**52.49.48.249**). Username: **ec2-user**. No local key pair needed for this connection method.

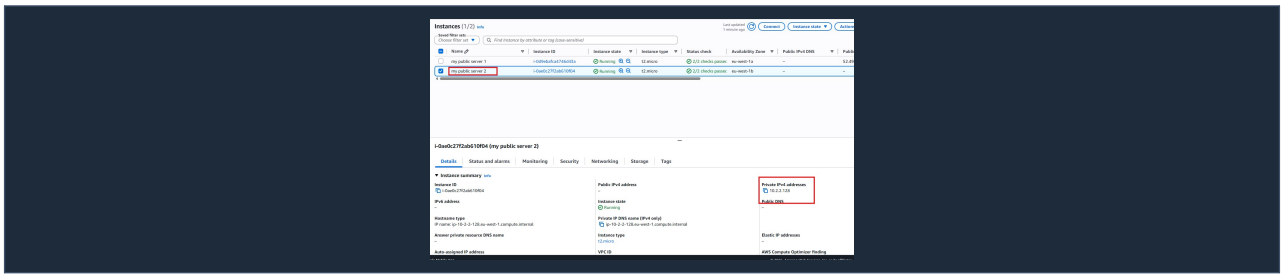


7 Validation — Ping Test

Cross-VPC Connectivity Confirmed

STEP 10 Server 2 Details — Private IP Only

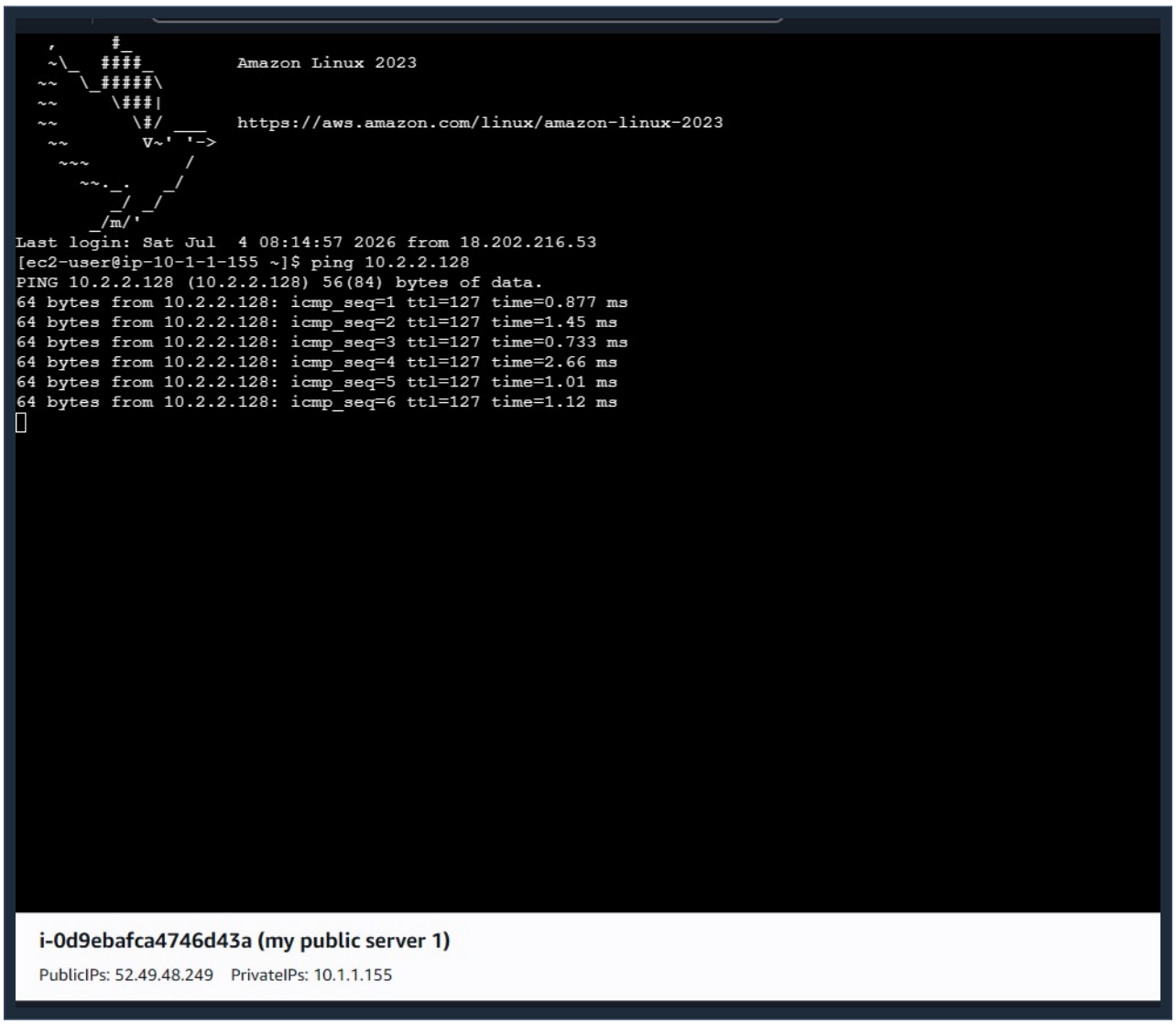
my public server 2 (i-0ae0c27f2ab610f04) has **Private IPv4: 10.2.2.128** and NO public IPv4 address. It runs in eu-west-1b. This confirms it is completely unreachable from the public internet — accessible only via the peering connection.



STEP 11 Ping Test: Server 1 → Server 2 Private IP (10.2.2.128)

From Server 1's terminal, **ping 10.2.2.128** was executed. ICMP replies confirm successful cross-VPC connectivity:

- VPC Peering Connection is routing traffic correctly
- Route tables on both VPCs direct peer traffic through the peering link
- Security Groups allow ICMP between the two instances
- Server 2 responds despite having no public IP or Internet Gateway



ping 10.2.2.128 | ttl=127 | ~0.8–2.6 ms | Connectivity Confirmed

Conclusion & Key Takeaways

Successfully established and verified a VPC Peering connection between two AWS VPCs — demonstrating that two completely separate network environments can communicate securely over AWS's internal backbone without traversing the public internet. VPC1 has full internet access via its Internet Gateway while VPC2 remains isolated, yet both servers communicate seamlessly through the peering link.

VPC Peering Basics

Allows two VPCs to communicate using private IPs without a VPN or Internet Gateway.

Route Table Updates

Both VPCs must update their route tables — pointing the peer CIDR to the pcx target.

One-Sided Internet Gateway

Only VPC1 has an IGW. VPC2 stays internet-isolated but is fully reachable from VPC1.

Security Group Control

SG1 allows SSH from anywhere (testing) and all traffic from VPC2's CIDR. SG2 allows from VPC1.

Elastic IP for Static Access

The Elastic IP on Server 1 gives a permanent public endpoint for SSH testing.

Validated, Not Assumed

The design was proven end-to-end with a live ping test — not just configuration checks.

AWS VPC Peer VPC | EC2 | Subnets | Route Tables | Security Grou Elastic IP | Internet Gateway